Tyler Katz

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EDUCATION

Syracuse University, School of Information Studies **Syracuse, NY**
Master of Science in Applied Human-Centered Artificial Intelligence *May 2027*
Syracuse University, School of Information Studies **Syracuse, NY**
Bachelor of Science in Applied Data Analytics *May 2026*

- **GPA:** 3.9/4.0; Dean's List (All semesters)

WORK & LEADERSHIP EXPERIENCE

Comcast **New York, NY (Remote)**
Data Science Intern *June - August 2026*

- Automated quarterly network device inventory reconciliation by building an LLM agent and MCP integration, cutting the process from hundreds of hours to 5 minutes and reducing token costs 30-40%.
- Engineered a FastAPI pipeline across 14 device platforms to auto-generate and upload CMDB templates, improving data consistency for ~5,000 network devices.
- Applied the agent to generate structured discrepancy reports across 14 platforms, cutting total reporting time from hundreds of hours to 2-3 hours.

United AI **Syracuse, NY**
Engineer Lead *September - December 2025*

- Sole engineer supporting 10 project teams of mostly non-technical members shipping working applications.
- Advised on Python, model training, and deployment, and reviewed designs for code quality across all 10 teams.
- Built the GitHub org and CI/CD (per-project write, org-wide read) and lectured 200+ members on Git and Claude Code.

Syracuse University **Syracuse, NY**
Undergraduate Data Analytics Research Assistant *January - December 2025*

- Engineered clean, integrated datasets from raw census files, geospatial shapefiles, and web-scraped sources using R, streamlining data preparation workflows and improving analysis efficiency for the project
- Designed and implemented data visualizations, geospatial maps, and statistical tables that were directly incorporated into multiple chapters of Machine [Gun] Politics, enhancing the clarity and persuasiveness of research findings
- Translated complex, multi-source datasets into actionable insights through advanced data wrangling, exploratory analysis, and visualization techniques, driving progress toward book completion and supporting evidence-based conclusions

PROJECTS

Fine-tuned LLM Microservice | *Python, QLoRA, Docker, FastAPI, AWS* | GitHub Repository *August 2026*

- Fine-tuned Qwen2.5-3B-Instruct via QLoRA (4-bit) for structured extraction of skills, tech stack, seniority, and comp from job postings, more than 8x'ing tech-stack F1 and lifting comp accuracy to 97% over the zero-shot baseline.
- Diagnosed a regressed fine-tune via raw-generation inspection, traced it to noisy data, and hit 100% valid-JSON on retrain.
- Containerized the fine-tuned model into a FastAPI service, deployed to AWS EC2 (Graviton/ARM) behind a public endpoint, and wired a GitHub Actions CI/CD pipeline to test, build, and push on every commit, plus latency-logging middleware.
- Cut the deployed container image from 4.26 GB to 475 MB by catching a CUDA-build default on a CPU-only target.

Agentic RAG System | *Python, LangGraph, Docker, FastAPI, Streamlit* | GitHub Repository *July 2026*

- Designed a 4-node LangGraph agent that decides when to retrieve, reformulates weak queries, and cites sources.
- Built an eval harness (hit-rate + LLM-as-judge) showing agent beats baseline: 90% vs 85% hit-rate, 2.80 vs 2.75 faithfulness.
- Shipped a Dockerized FastAPI + Streamlit RAG service answering questions over 100 job postings and 250 papers.

Inventory Reconciliation Agent - Comcast (Internal Tool) | *Python, LLM Agent, MCP Server* *June - July 2026*

- Built an LLM reconciliation agent on a new in-house platform, one of the first use cases to demonstrate its effectiveness.
- Redesigned the agent's architecture from an executable skill (script run and summarized in-platform) to an instructional skill (script run externally, output uploaded with structured formatting instructions), cutting token usage 30-40%.
- Developed a custom MCP integration enabling the agent to autonomously remediate its own reconciliation recommendations.

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages: Python | SQL | R | JavaScript
AI & ML: LangGraph | RAG | Chroma | Pinecone | QLoRA/PEFT | PyTorch | Hugging Face | scikit-learn | XGBoost | MCP
Frameworks & Tools: FastAPI | Docker | AWS (EC2) | GitHub Actions CI/CD | Git | pytest | PySpark | Tableau | Power BI